Basic Calculator (Hard)

Problem Statement

Given a string `s`, representing the `arithmetic expressions`, evaluate the expression and return the resultant value.

Constraints:

- s contains digits, '+', '-', '(', ')', and ' '.
- '+' is used as addition operation but not as a unary operator. (i.e., "+2" and "+(1 + 3)" is invalid).
- '-' is used as a unary and subtraction operator both. (i.e., "-2" and "-(1 + 3)" is valid).

Note: You are not allowed to use the built-in function like `eval()` to evaluate the mathematical expression.

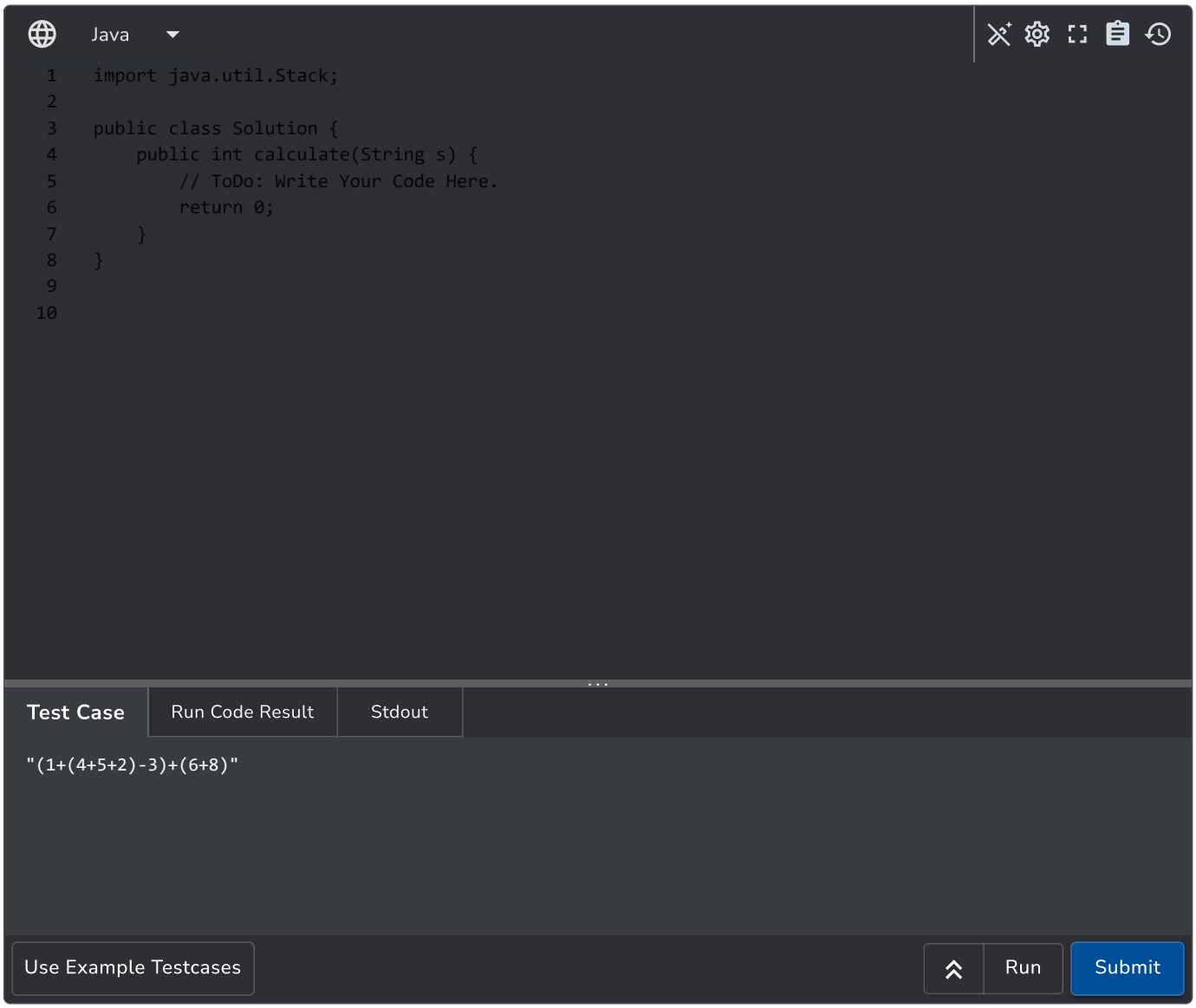
Examples

- **Example 1**:
  - **Input**: `"3 - (2 - 1) + 4"`
  - **Expected Output**: `6`
  - **Justification**: The expression inside the parentheses `2 - 1` evaluates to `1`. So the expression becomes `3 - 1 + 4`, which sums up to `6`.
- **Example 2**:
  - **Input**: `"10 + (8 - (4 + 2)) - 1"`
  - **Expected Output**: `11`
  - **Justification**: Inside the inner parentheses, `4 + 2` equals `6`. The expression becomes `10 + (8 - 6) - 1`. Then `8 - 6` equals `2`, so it simplifies to `10 + 2 - 1`, which is `11`.
- **Example 3**:
  - **Input**: `"5 + (3 - (2 + (8 - 5)) + 7) - (6 - 4)"`
  - **Expected Output**: `8`

- **Justification**: In the innermost parentheses, `8 - 5` equals `3`. The expression becomes `5 + (3 - (2 + 3) + 7) - (6 - 4)`. The next inner expression `2 + 3` equals `5`, so it becomes `5 + (3 - 5 + 7) - 2`. Simplifying further to `5 + 5 - 2`, we get `10 - 2`, which is `8`.

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